



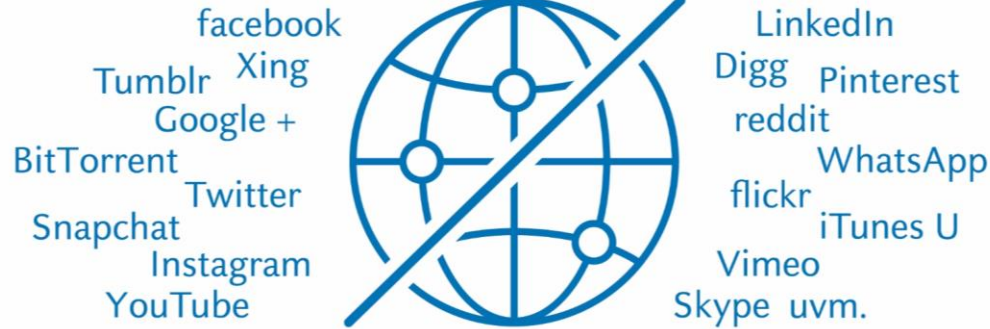
MoPoS – Makromolekulare Chemie II

SoSe 2023



Bitte verzichten Sie in dieser Lehrveranstaltung
auf Bild- und Tonaufzeichnungen.

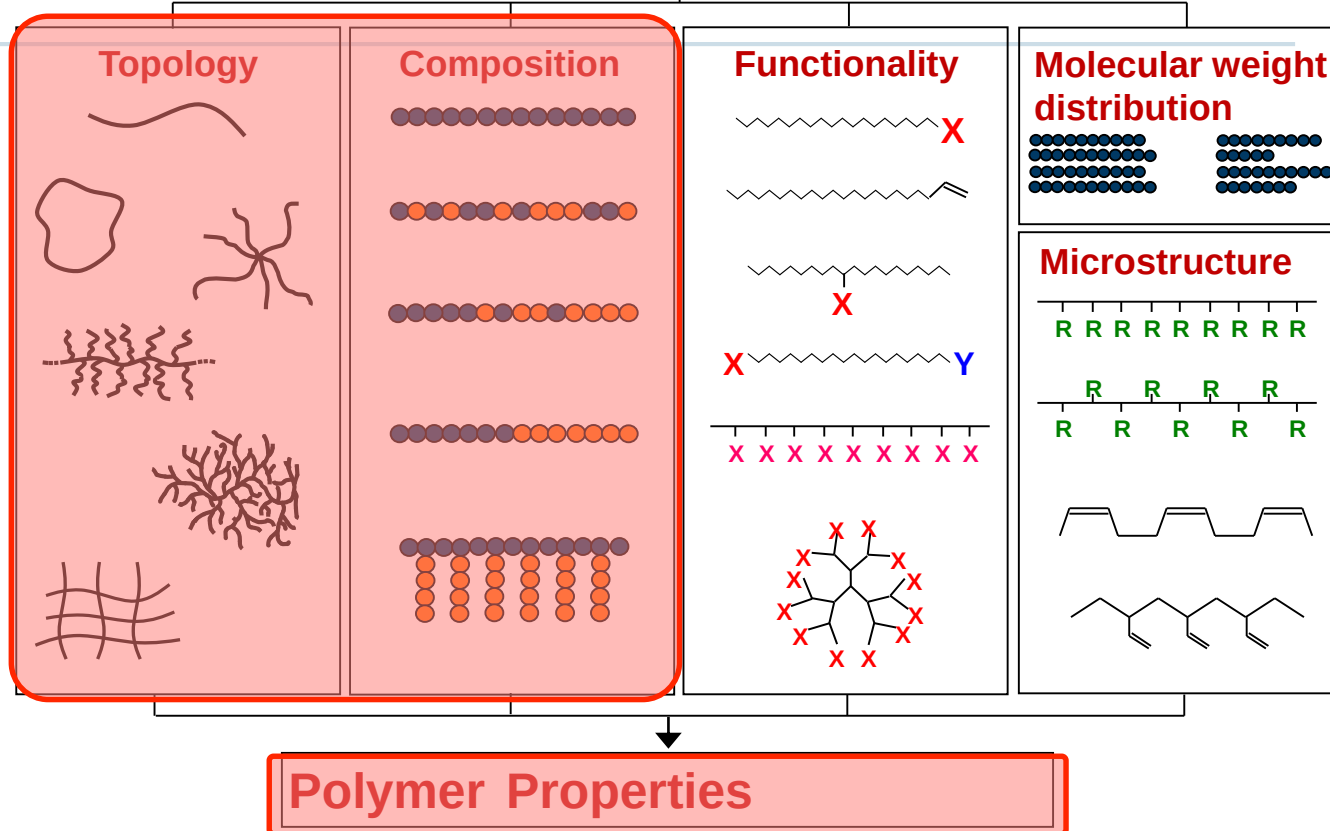
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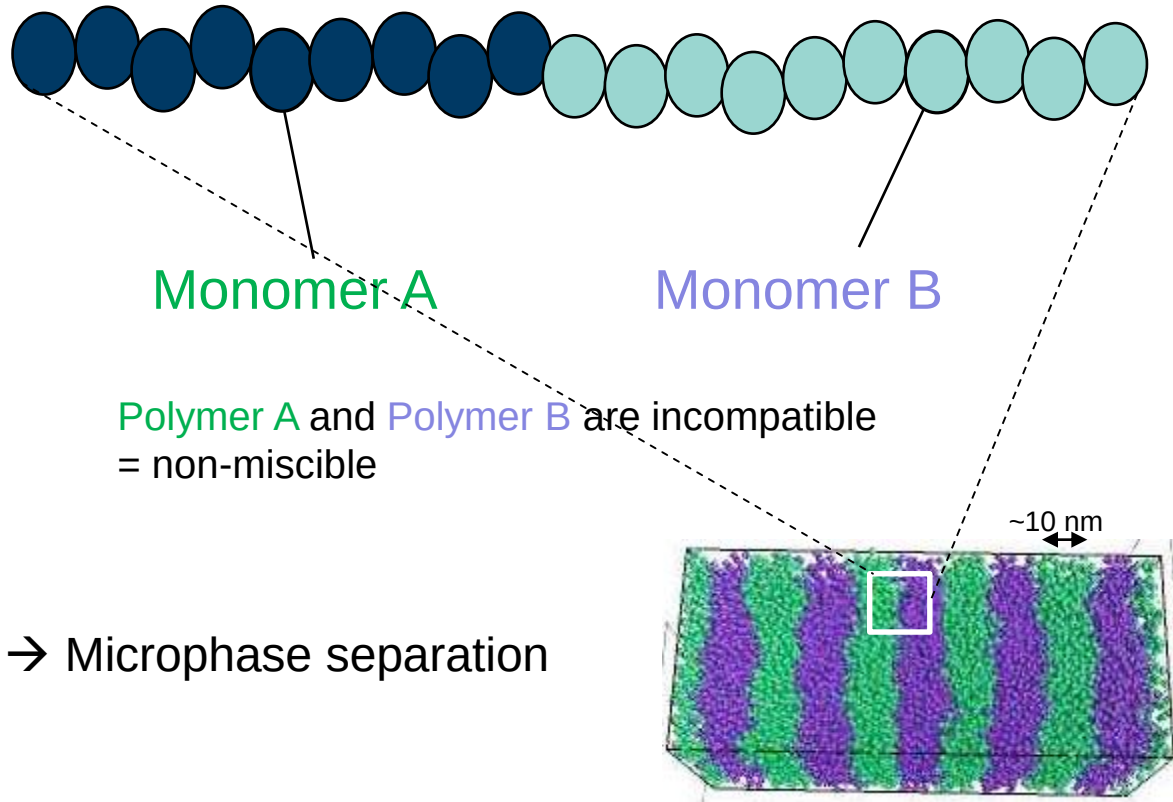
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Tonaufzeichnungen dieser Lehrveranstaltung
ist nicht gestattet.

Weitere Informationen finden Sie im Studierendenportal

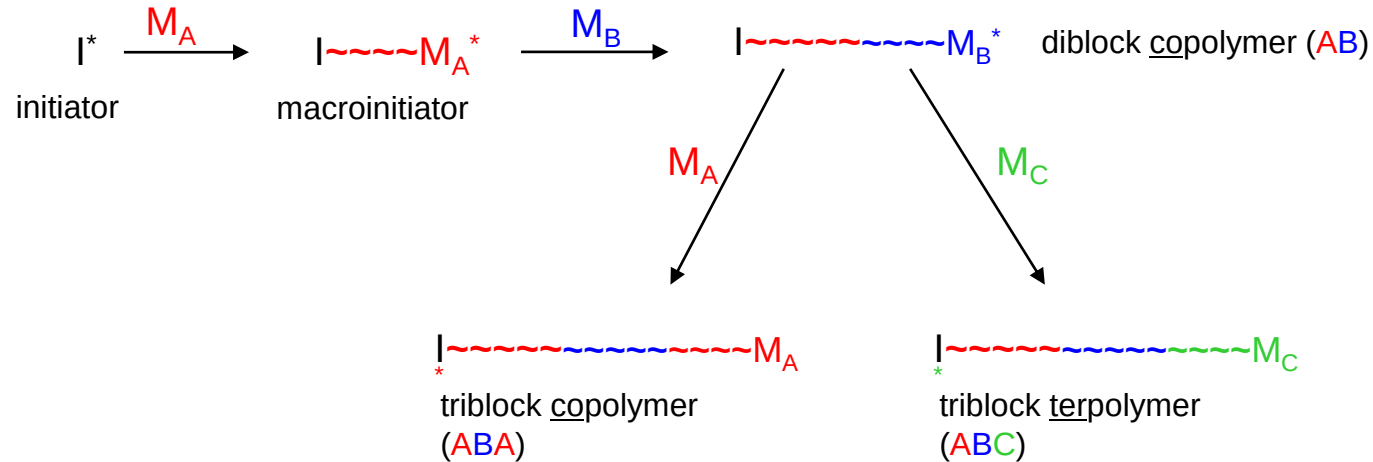
Polymer Architecture



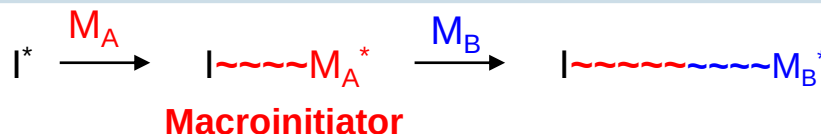
Block Copolymers



Synthesis of Block Copolymers by Sequential Monomer Addition



Sequential Monomer Addition



Good blocking efficiency in anionic polymerization when the nucleophilicity of the first block is higher or comparable to that of the second block.

Monomer Type	pK _a (DMSO)	pK _a (H ₂ O)	Initiators ^b
Ethylene	56		RLi
Dienes and Styrenes	44		NR ₂ ⁻ , RLi,
	43		RMt ^b , naphthalene radical anions ^c , cumyl ⁻ K ⁺ , Mt,
Acrylonitrile	32		RMgX
Alkyl	30–31 (195)	27–28 (195)	Fluorenyl ⁻ ,
Methacrylates			Ar ₂ C ⁻ , ketyl radical anions ^d
Vinyl ketones	26	19 (197)	
Oxiranes	29–32	16–18 (198)	RO ⁻
Thiiranes	17	12–13 (199)	
Nitroalkenes	17	10–14 (200)	
Siloxanes		10–14 (201, 202)	RO ⁻ , OH ⁻
Lactones	12	4–5 (203)	RCO ₂ ⁻
Cyanoacrylates	11 (196)		HCO ₃ ⁻ , H ₂ O
Vinylidene cyanide	11	11 (196)	

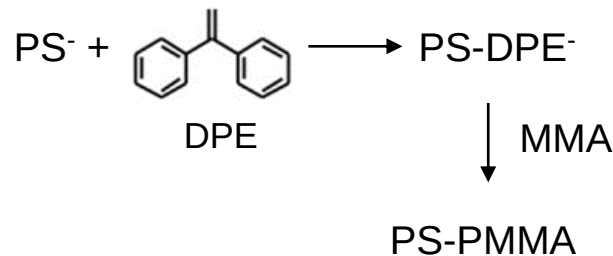
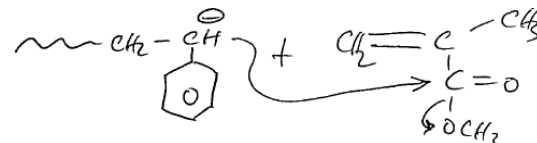
^apK_a values refer to the conjugate acid of the anionic propagating intermediate. pK_a values in DMSO are from Ref. 194 unless noted in parentheses after the number. The references for pK_a values in H₂O are listed in parentheses after the number.

^bMt refers generally to alkali metals (Li, Na, K, Rb, Cs).

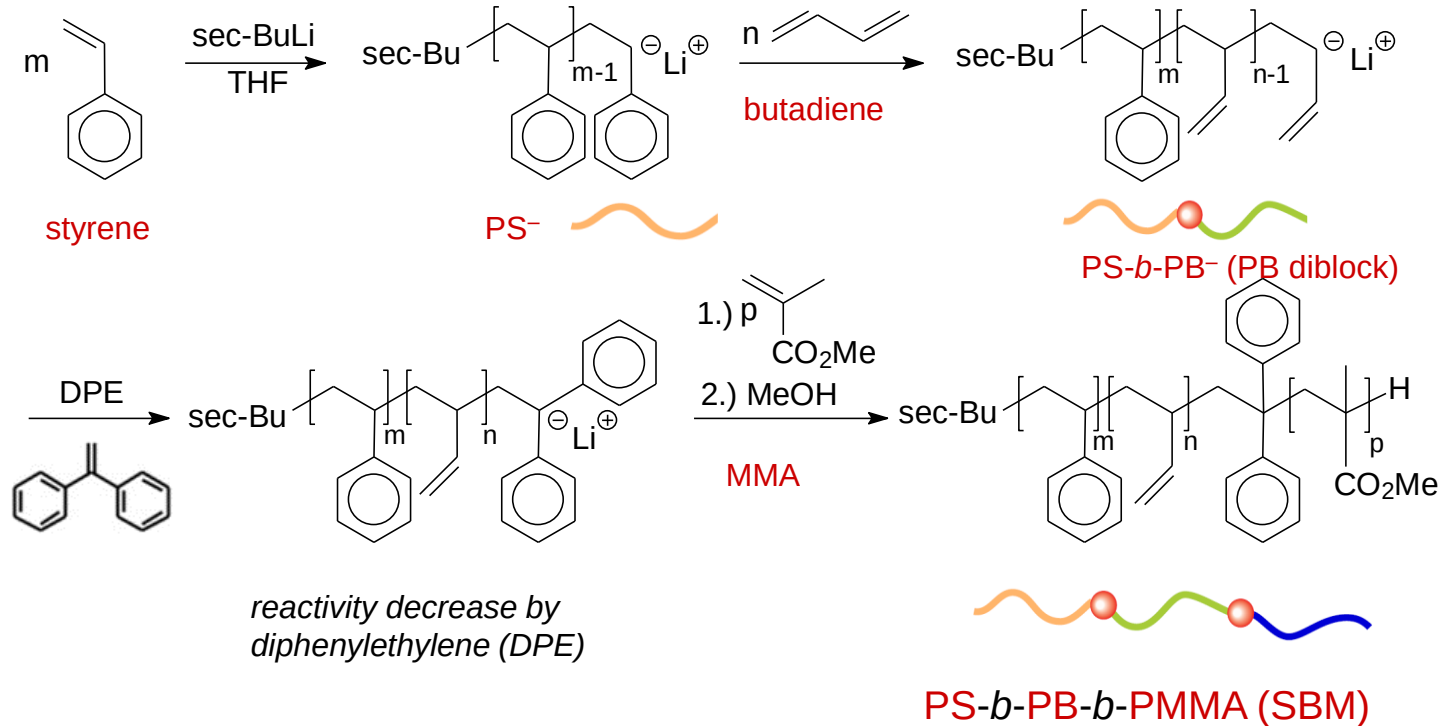
^cFor example, naphthalene radical anion ⁻ (Li⁺, Na⁺, K⁺).

^dAr₂CO⁻.

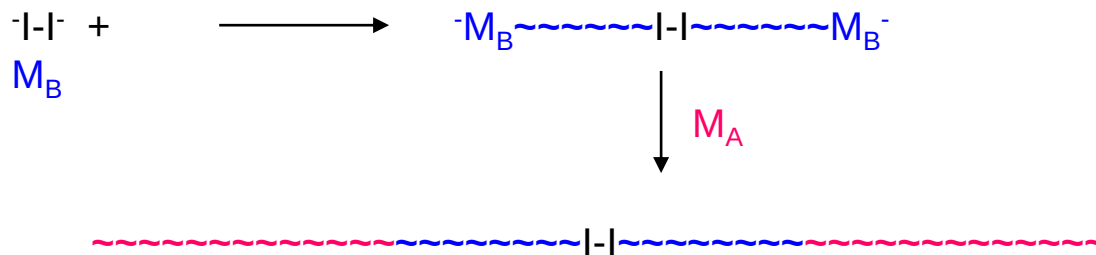
Decreasing the reactivity of the first block may be necessary in order to avoid side reactions in the cross-over step.



Synthesis of Poly(styrene-*block*-Poly(butadiene-*block*-PMMA (SBM) Triblock Terpolymer via Anionic Polymerization



ABA Triblock Copolymers by Bifunctional Initiation



Problem: **Bifunctional initiators are not well soluble in hydrocarbon solvents (due to aggregation)**

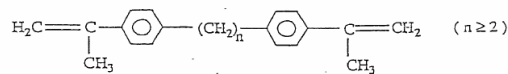
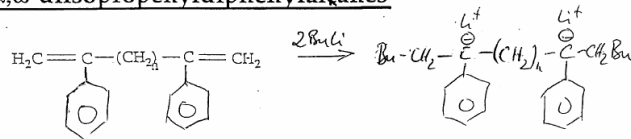
Most important **properties**: Microphase separation in bulk
Micelle formation in solution

Most important **applications**: SBS, SIS: thermoplastic elastomers (Kraton®)
PEO-PPO-PEO, amphiphilic copolymers (Pluronic®)

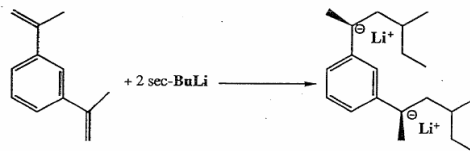
Bifunctional initiators for non-polar solvents

1) α,ω -bis(phenylvinylidenyl)alkanes

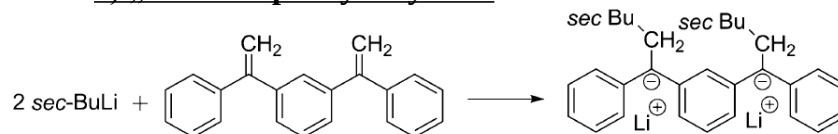
α,ω -diisopropenyldiphenylalkanes



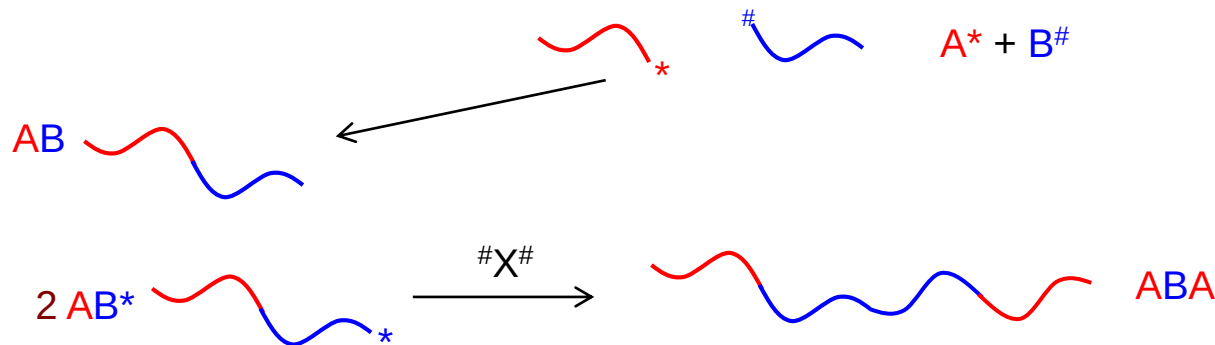
2) Diisopropenylbenzenes



3) „double diphenylethylene“



Synthesis of Block Copolymers by Coupling



Example:

* = PS-Li, # = Br-CH₂-Phenyl

Problems:

low coupling efficiencies

stoichiometry is difficult

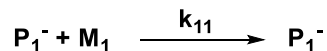


- Special case: reaction between living anion and living cation:



Block copolymers by statistical copolymerization?

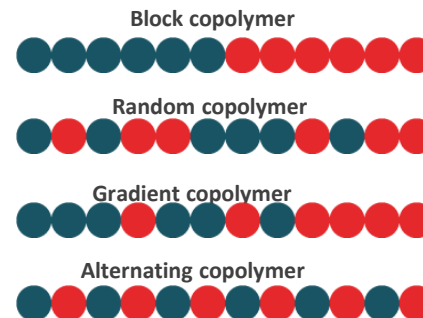
-> Living Copolymerization



$$r_1 = \frac{k_{11}}{k_{12}}$$

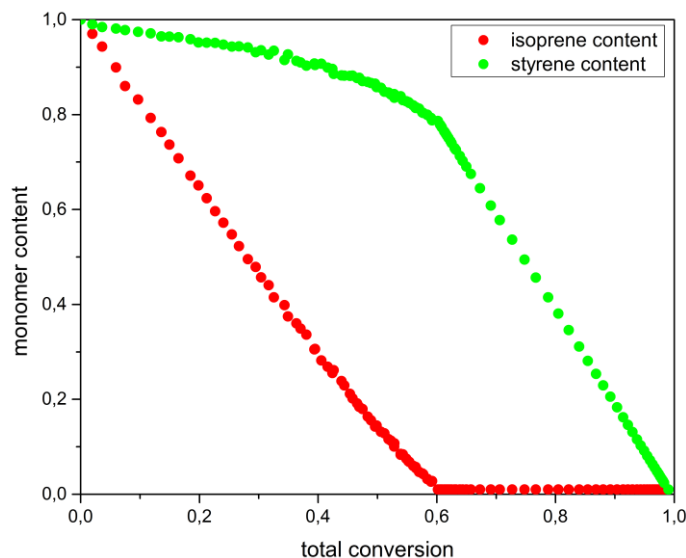
$$r_2 = \frac{k_{22}}{k_{21}}$$

$$r_1 > r_2 \rightarrow$$



$r_1 \gg r_2$ and $r_2 < 1 \rightarrow$ Tapered block copolymer



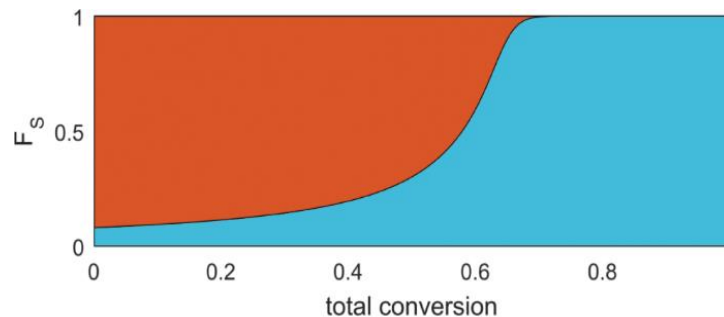


Random anionic
copolymerization
in cyclohexane

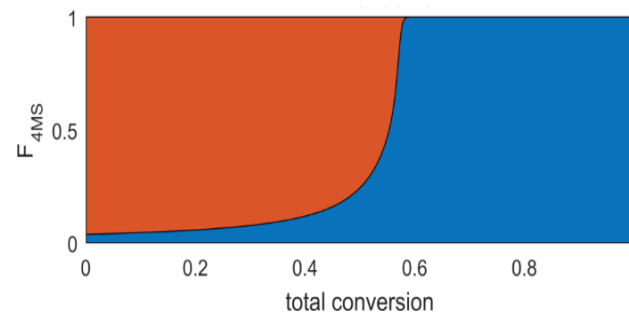
Method	r_{isoprene}	r_{styrene}
Fineman Ross	11.21	0.058
Kelen Tüdös	10.71	0.049
Meyer-Lowry	11.07	0.040
Average Value	11.0	0.049
Literature Value ¹	14.4	0.045
Literature Value ²	12.8	0.051

Shape of the Gradient

I / S:
 $r_I = 11.0$
 $r_S = 0.053$



I / 4-MS:
 $r_I = 25.4$
 $r_{4MS} = 0.007$



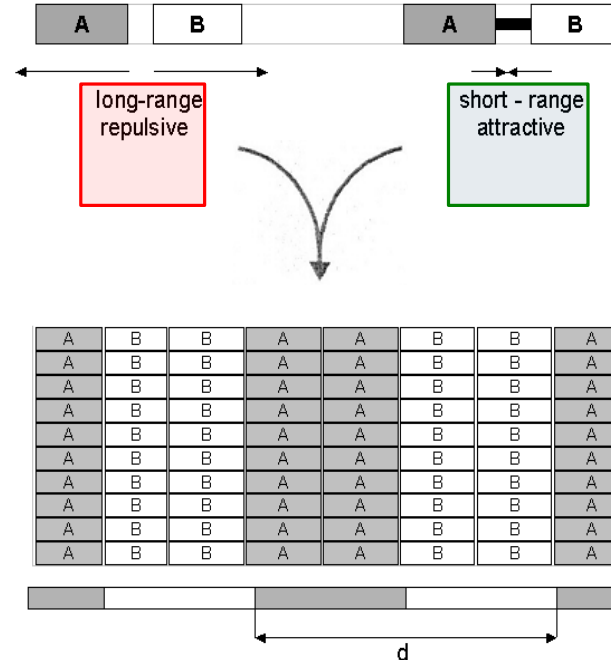
Properties of AB Diblock and ABA Triblock Copolymers and of ABC Triblock Terpolymers

- A system produces a property it did not have before (order)
- Transition from disordered to ordered phase
by a change in thermodynamic or physical property :
 - temperature, concentration, pH
 - derived property (polarity, degree of substitution)
 - mechanical field (pressure, ultrasound, shear)
 - electric, magnetic field

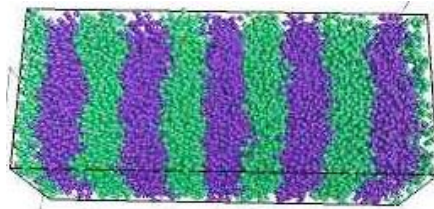
Molecular Prerequisites for Self-Ordering

Simultaneous existence of

- long-range repulsive force: incompatibility
- short-range attractive force: covalent bond



Block copolymer morphologies are linked → microphase separation → special morphologies

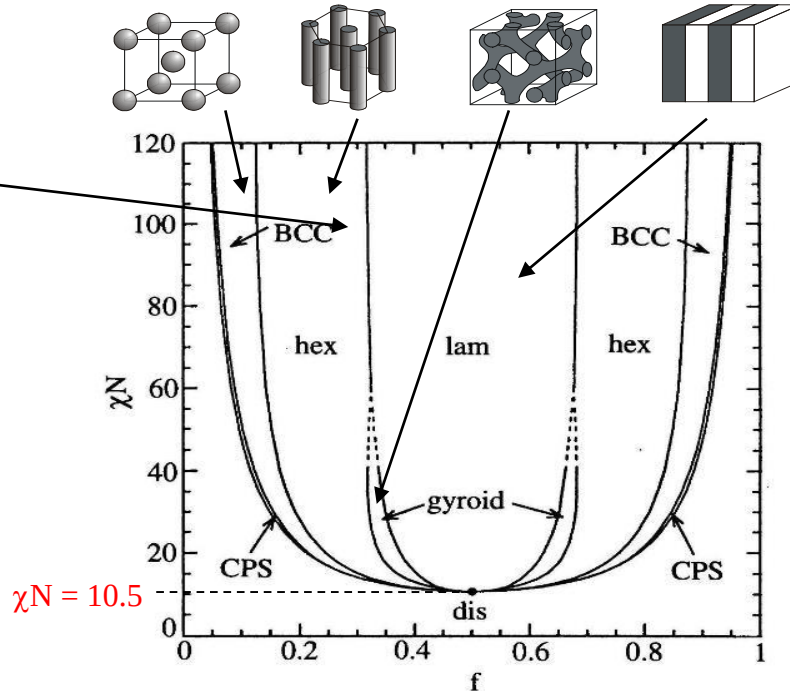


Calculated phase diagram
for AB diblock copolymers:
BCC = body-centered cubic
packed spheres
CPS = cubic packed spheres
hex = hexagonal cylinders
lam = lamellae
dis = disordered

χ = Flory-Huggins parameter

N = degree of polymerization

f = volume fraction of block 1



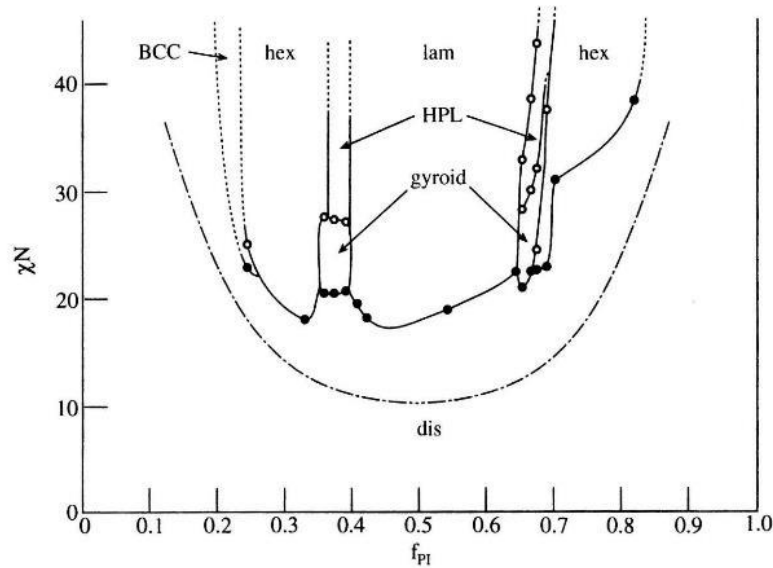
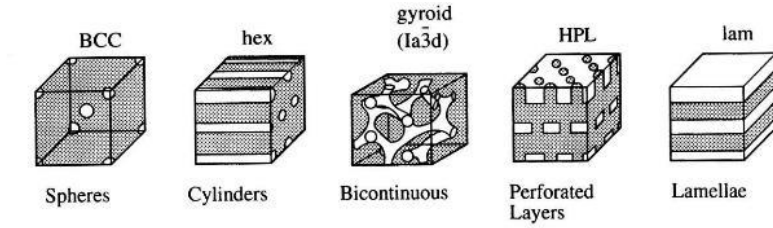
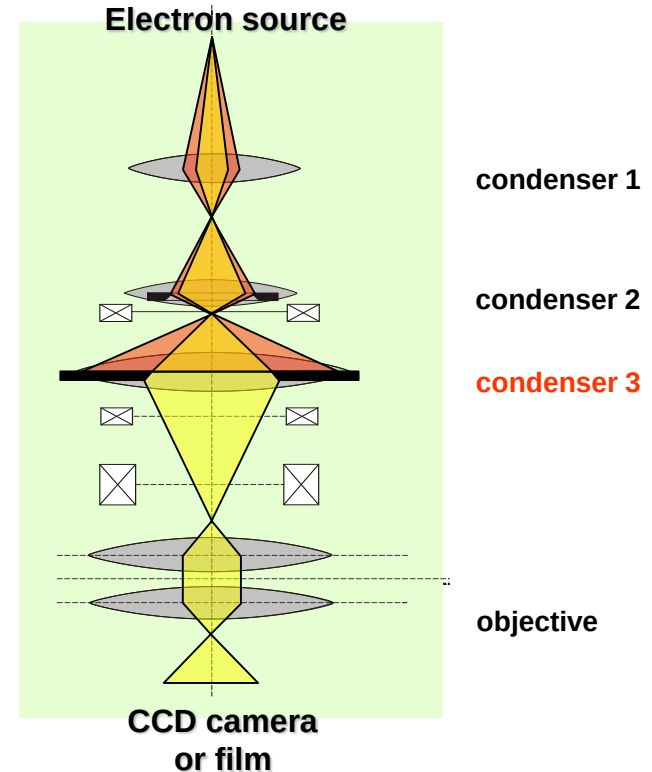


Fig. 2.2 Experimentally-determined phase diagram for PS-PI diblock copolymers (Khandpur *et al.* 1995).

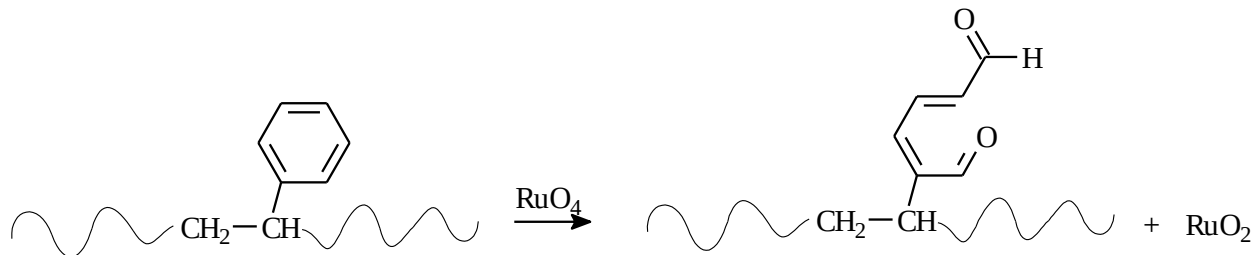
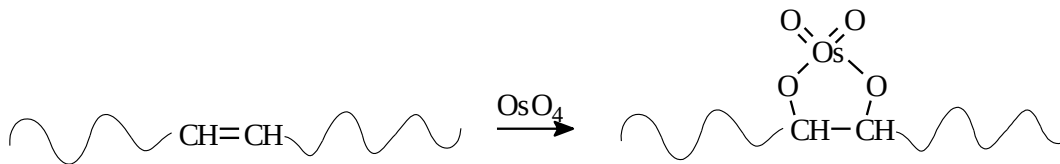
The Transmission Electron Microscope (TEM)



Resolution down to $\text{\AA}$ scale!

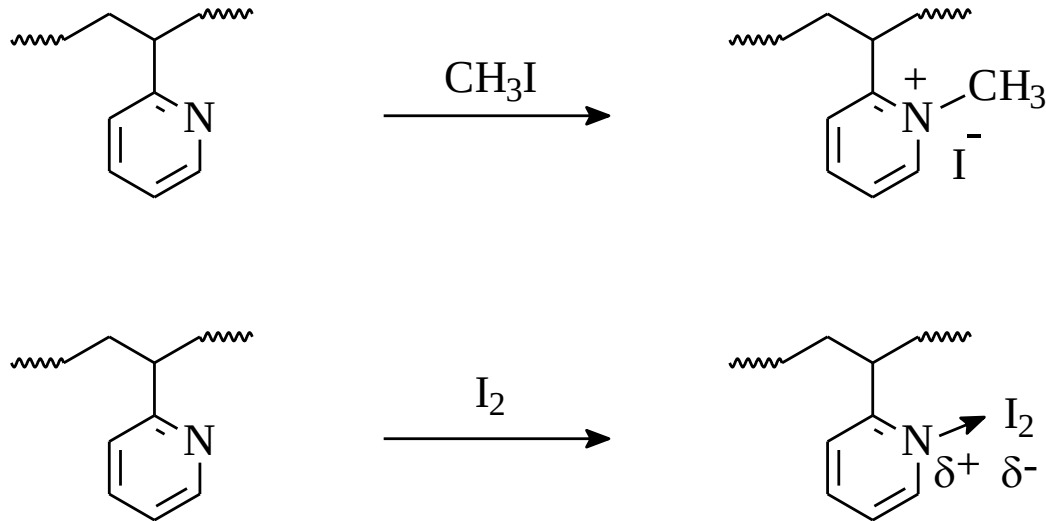


1. Cast film from solution (very slow evaporation)
2. Stain with heavy atom compounds:



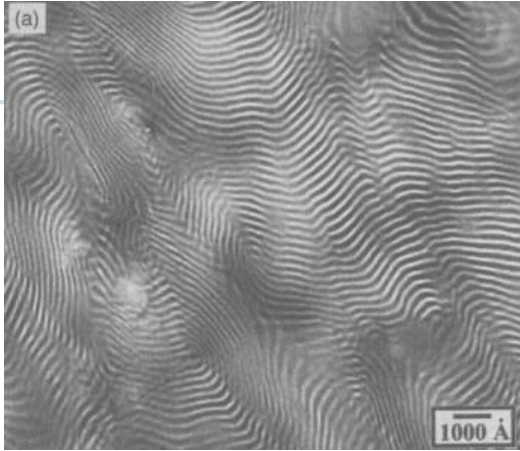
3. Cut thin (20-50 nm) slices using a microtome

Staining with heavy atom compounds (2):



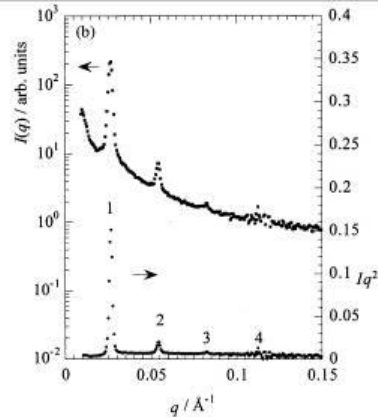
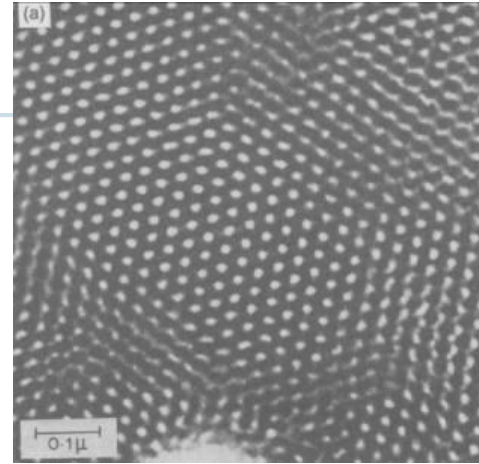
3. Cut thin (20-50 nm) slices using a microtome

lamellar morphology



TEM

cubic (spherical) morphology



Small-Angle
X-Ray
Scattering
(SAXS)

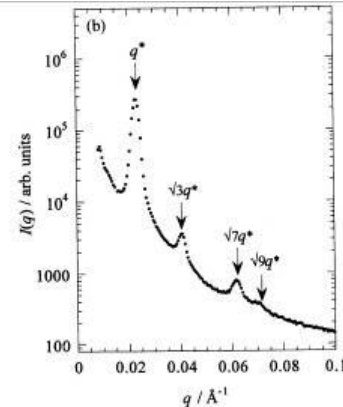
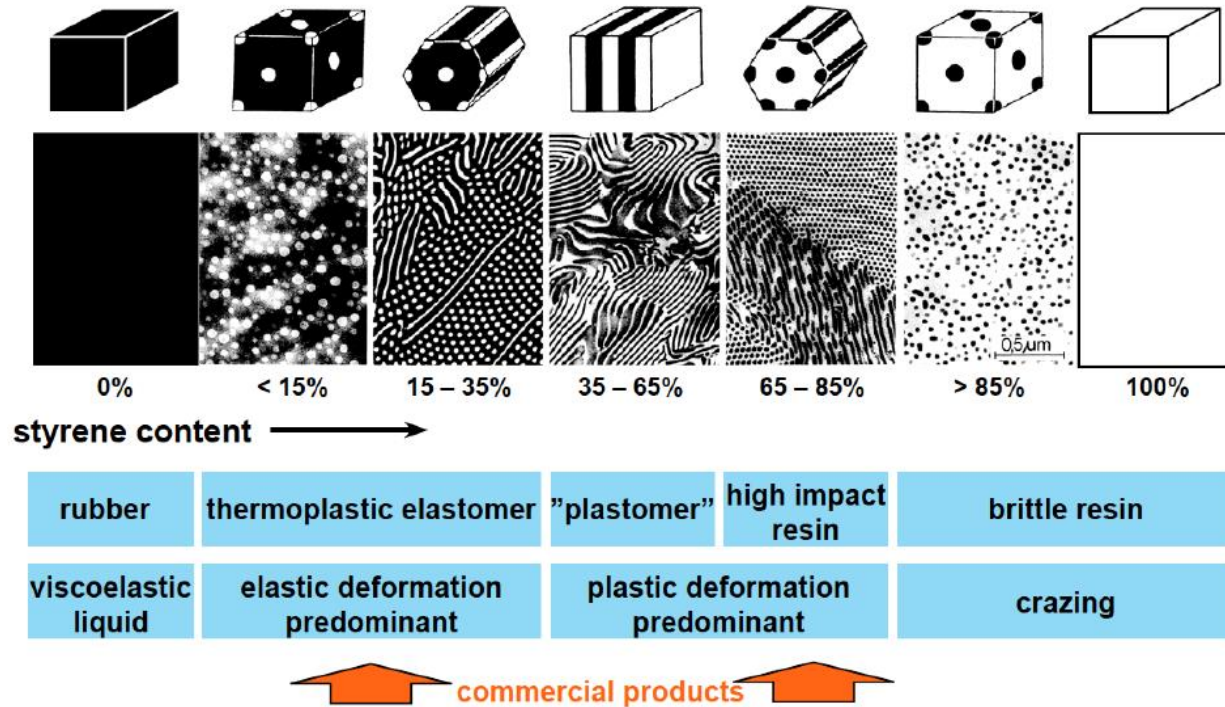


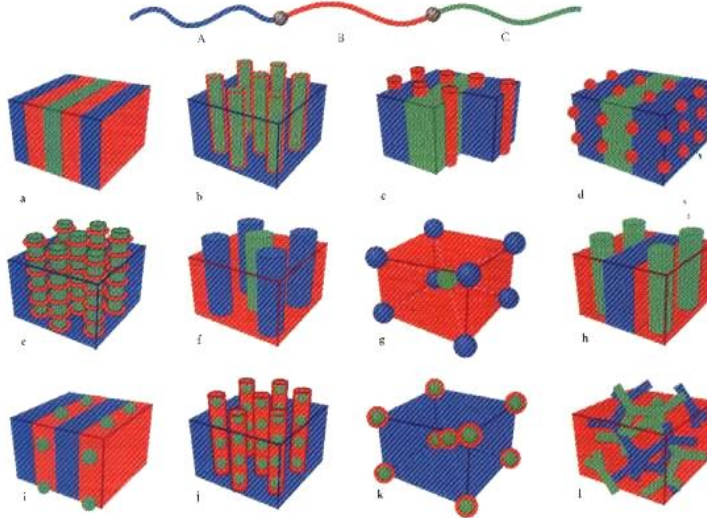
Fig. 2.6 (a) The lamellar microstructure of a PS-PI diblock with $f_{PS} = 0.64$ and $M_n = 39$ kg mol⁻¹ shown by TEM where the PI is stained dark (Khandpur *et al.* 1995); (b) SAXS powder pattern from the same material.

Fig. 2.9 (a) TEM micrographs from the hexagonal morphology in Kraton TR1102, a PS-PB-PS block copolymer with $f_{PS} \approx 0.24$ (Dlugosz *et al.* 1970; Keller and Odell 1985). (b) SAXS powder pattern from Kraton TR1102 (Ryan and Hamley 1997).

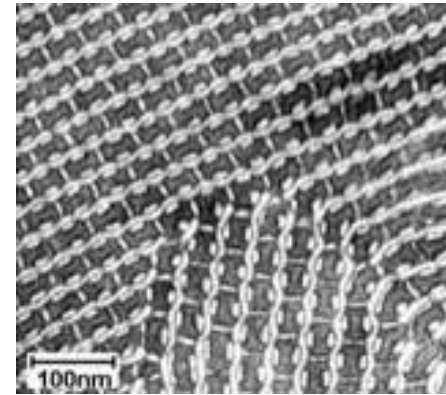
SB and SBS Block Copolymers



Bulk morphologies of ABC triblock terpolymers (ABC block copolymers)

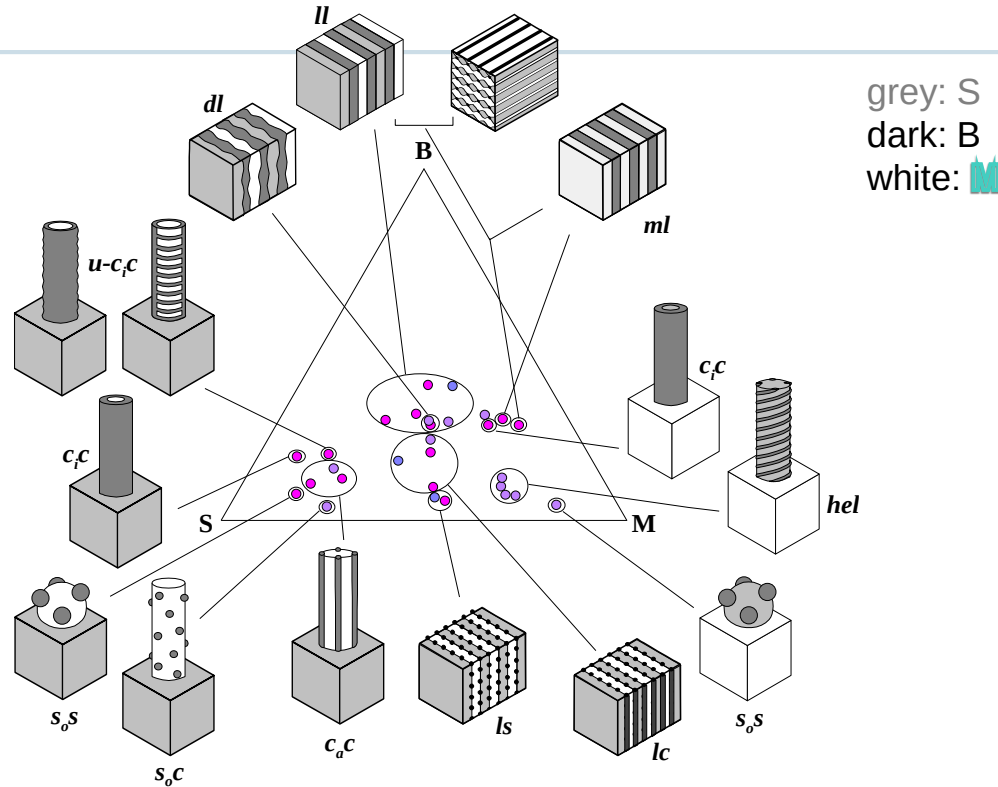


“knitting pattern”

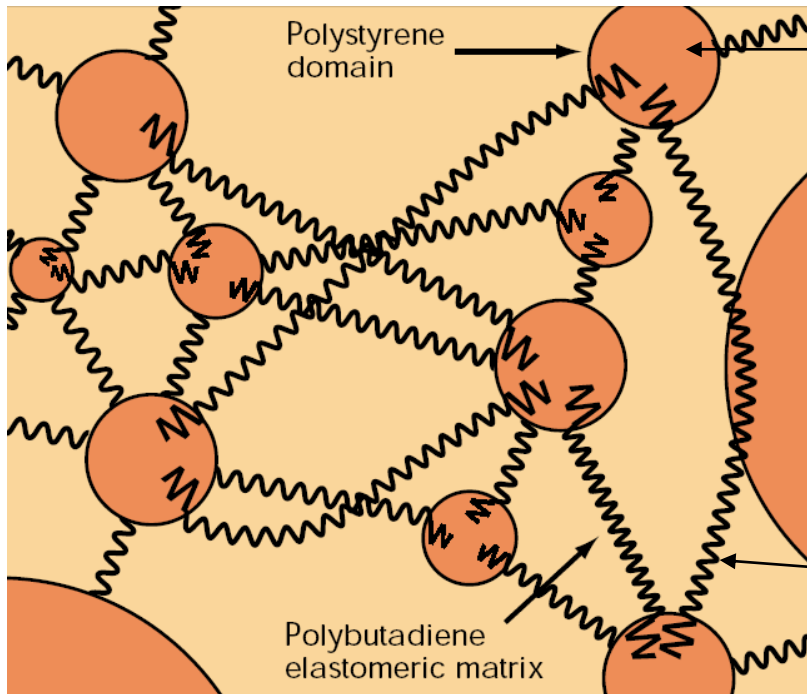


Stadler et al.

Morphologies in PS-PB-PMMA (SBM) triblock copolymers



Applications: Thermoplastic Elastomers



PS block is glassy below 100 ° C,
but can be processed like a plastic
above

PS-b-PB-b-PS (SBS)

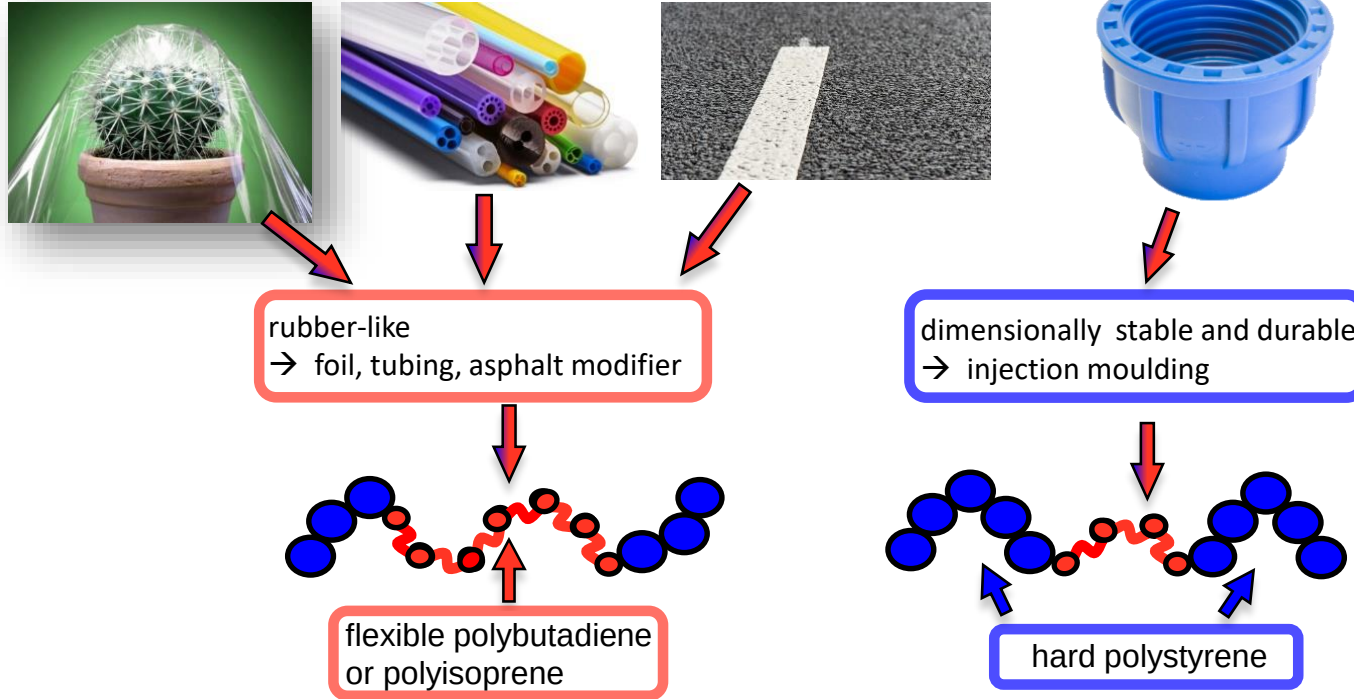
the rubbery PB chains are
physically crosslinked by the
PS spheres

Trade name, e.g. **Kraton™**

Use for adhesives, recycleble shoe soles, stretch wrapping film, additives for asphalt, etc.



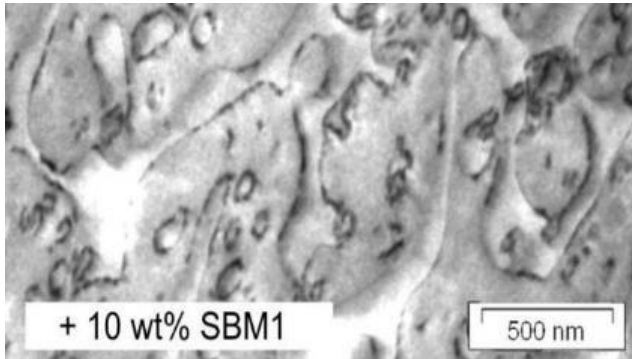
SBS and SIS Triblock Copolymers: TPEs



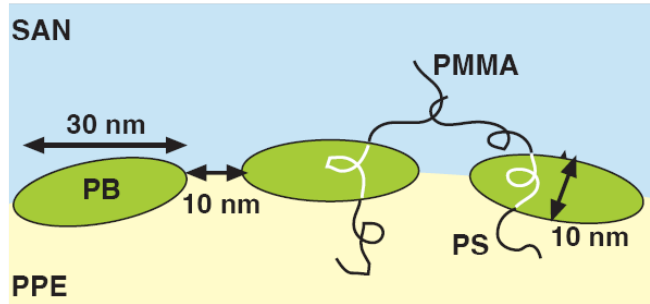
→ Synthesis by Sequential Anionic Polymerization

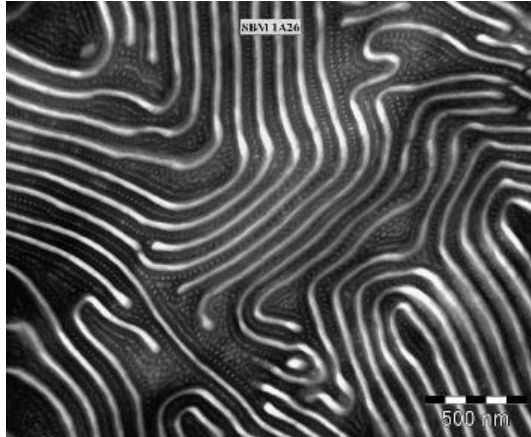
PS-*b*-PB-*b*-PMMA (SBM) as Compatibilizer for PPE/SAN Blends

■ PPE/SAN = Poly(phenylene ether)/Poly(styrene-co-acrylonitrile)

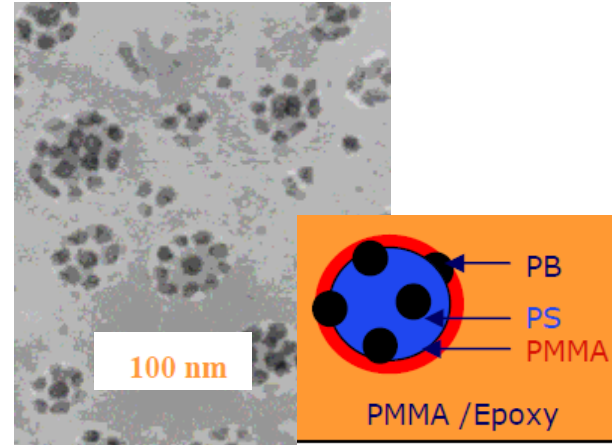


PPE/SAN 60/40
+ 10% SBM





*Bulk morphology of SBM
(TEM)*



*Nanostructuring of an
epoxy resin modified
with SBM (TEM)*

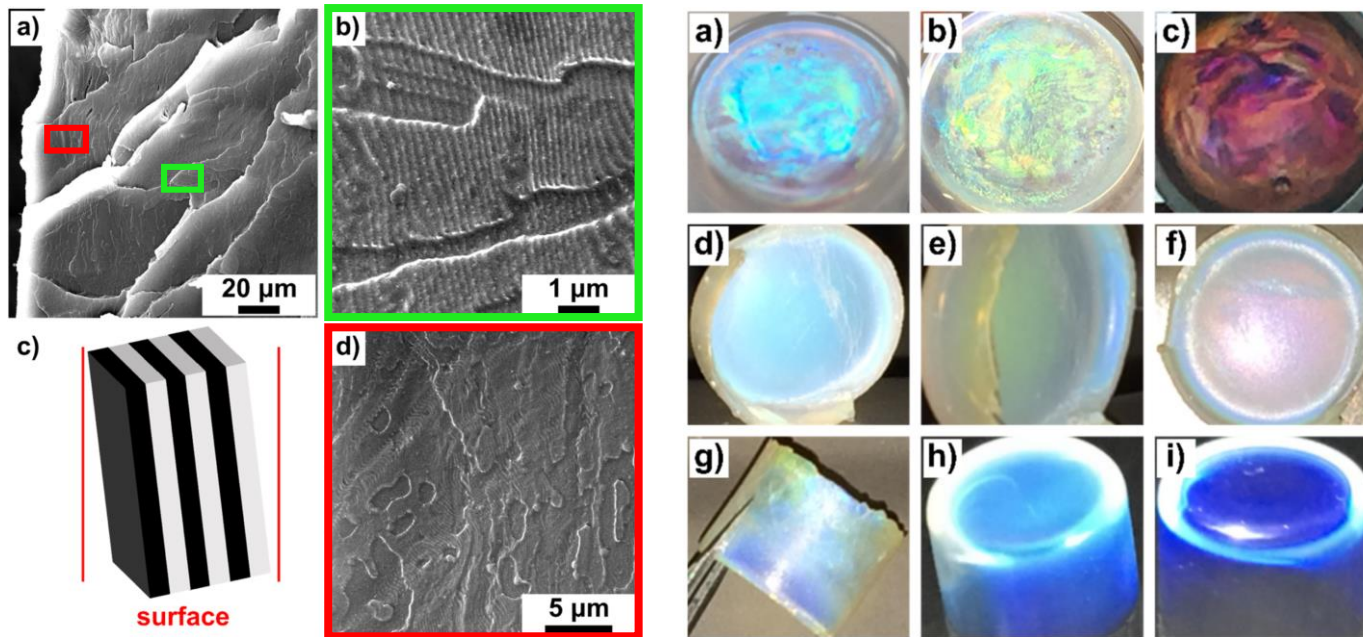


NANOSTRENGTH® Thermoset Applications



- Toughening with Nanostrength®
 - Tough, rigid, high Tg materials, transparency

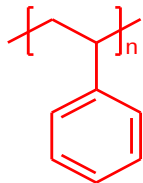
Ultra high Molecular weight Block Copolymers (PI-b-P4MS)



domain sizes: 100–200 nm

vivid structural colours

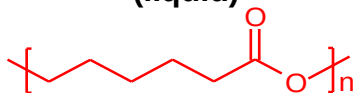
Hydrophobic Blocks



Polystyrene (glassy)

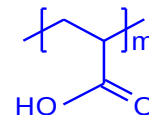


**Polybutadiene
(liquid)**

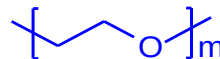


**Polycaprolactone
(crystalline)**

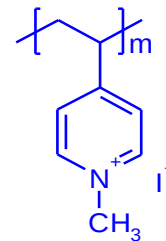
Hydrophilic Blocks



Poly(acrylic acid) (weak polyanion)

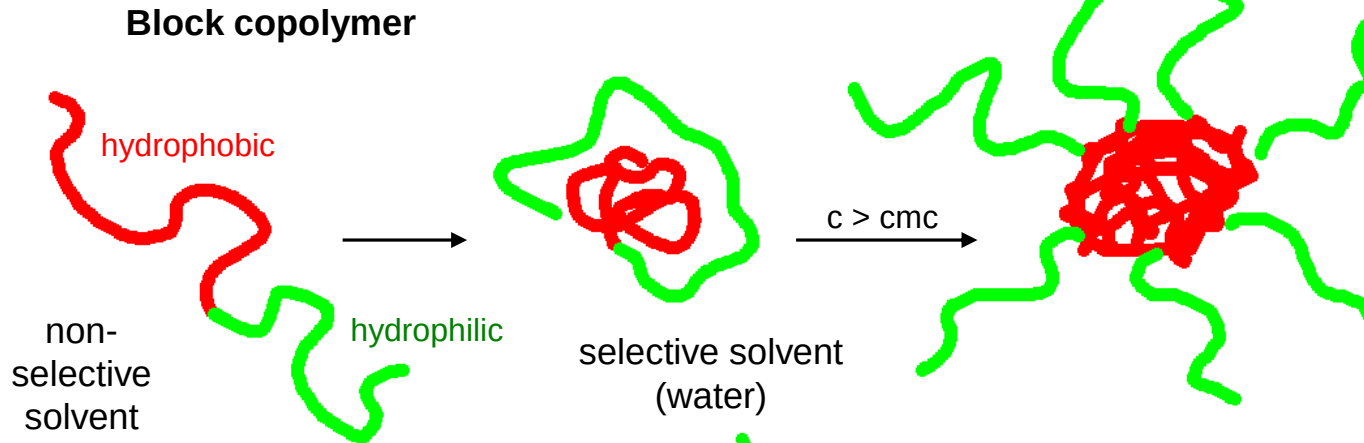


Poly(ethylene oxide) (neutral)



Poly(4VPQ) (strong polycation)

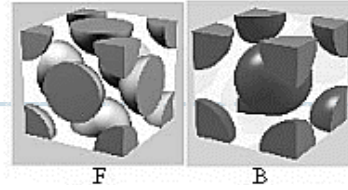
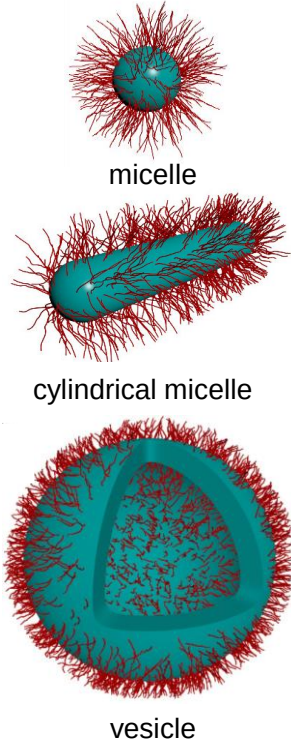
Micelle Formation of Amphiphilic Copolymers



Some applications:

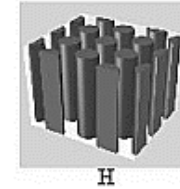
- dispersants for cosmetics, pigments, or latex particles
- carriers for drugs, enzymes, DNA, nanoparticles
- removal of pollutants
- associative thickeners

Solution Structures of Block Copolymers in a Selective Solvent

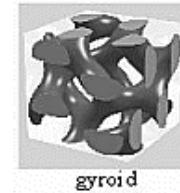


spheres

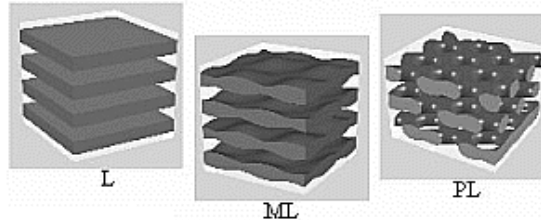
Bulk Structures of Block Copolymers



cylinders



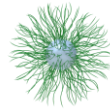
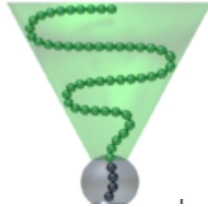
gyroids



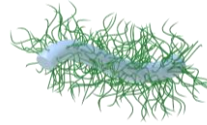
lamellae

Dependence of Morphologies on Corona Volume

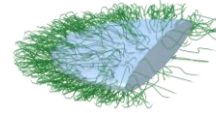
AB
Diblock



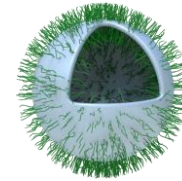
Sphere



Cylinder

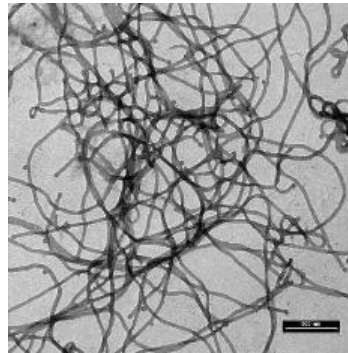
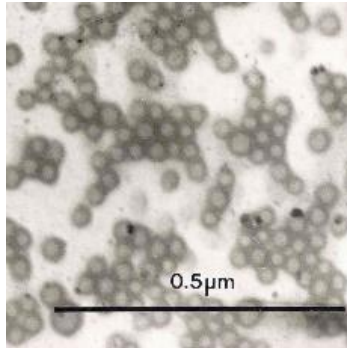


Bilayer sheet

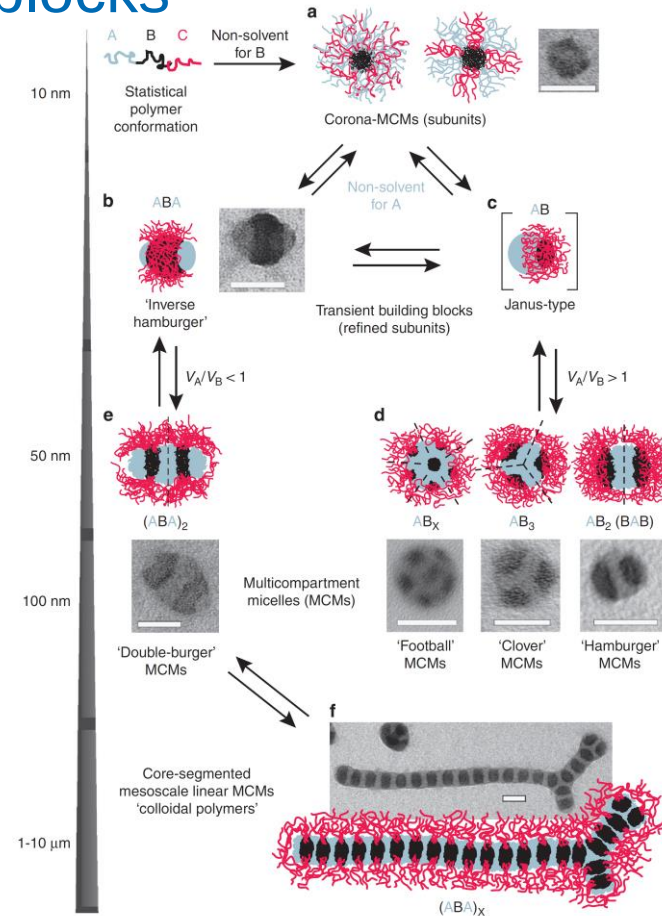
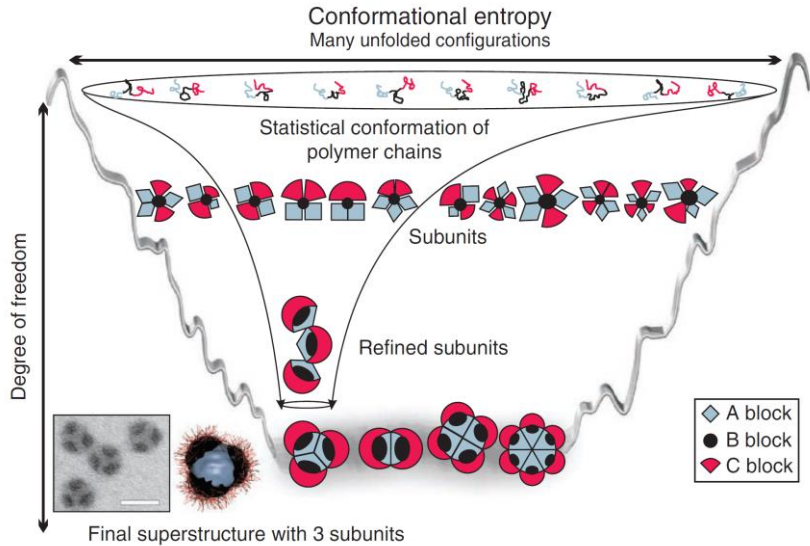


Vesicle
(Polymersome)

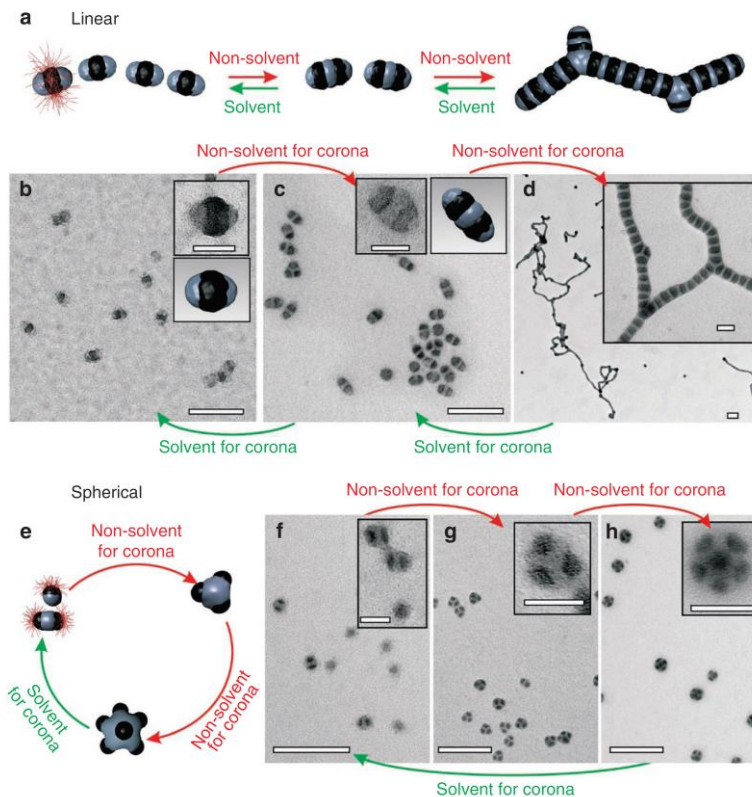
decreasing corona
volume



Complex Architectures from Triblocks



Complex Architectures from Triblocks



1. Critical Micelle Concentration (cmc):

- Surface Tension (Tensiometry)
- Fluorescence Spectroscopy (Fluorometry)
- Static and Dynamic Light scattering (SLS, DLS)
- NMR

2. Aggregation number, radius of gyration:

- Static Light Scattering (SLS)
- Small-Angle Neutron/X-ray Scattering (SANS, SAXS)

3. Size (hydrodynamic radius):

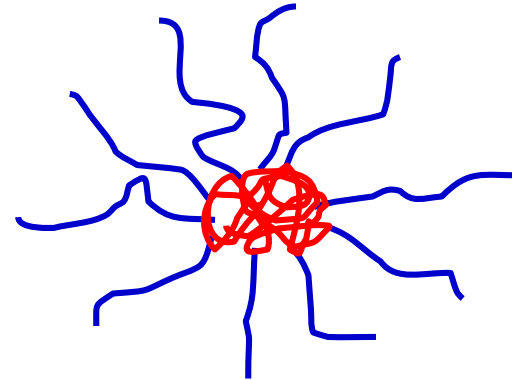
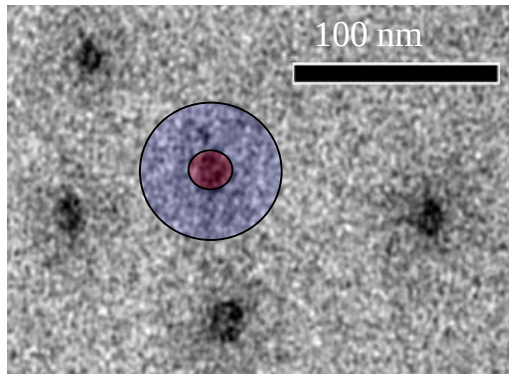
- Dynamic Light Scattering (DLS)
- Field Flow Fractionation (FFF)
- Analytical Ultracentrifugation (AUC)

4. Shape:

- Radius of gyration vs. hydrodynamic radius (R_g/R_h)
- SANS, SAXS
- Transmission Electron Microscopy (TEM)
- Scanning Force/Electron Microscopy (AFM, SEM)

Formation of spherical micelles: Cryo-TEM

PnBA₉₀-PAA₃₀₀

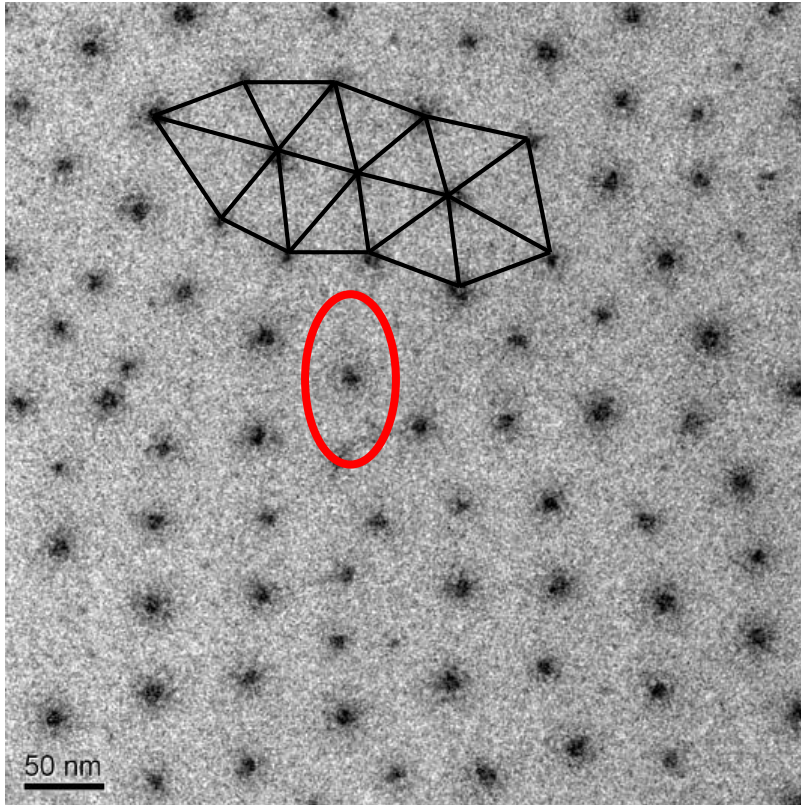


$R_{\text{core}} \sim 5 \text{ nm}$

$R_{\text{micelle}} \sim 30 - 50 \text{ nm}$

Collapsed core surrounded by **an extended corona**

Micelles of Polyisobutylene-b-PMAA



PIB₃₁-PMAA₁₇₁

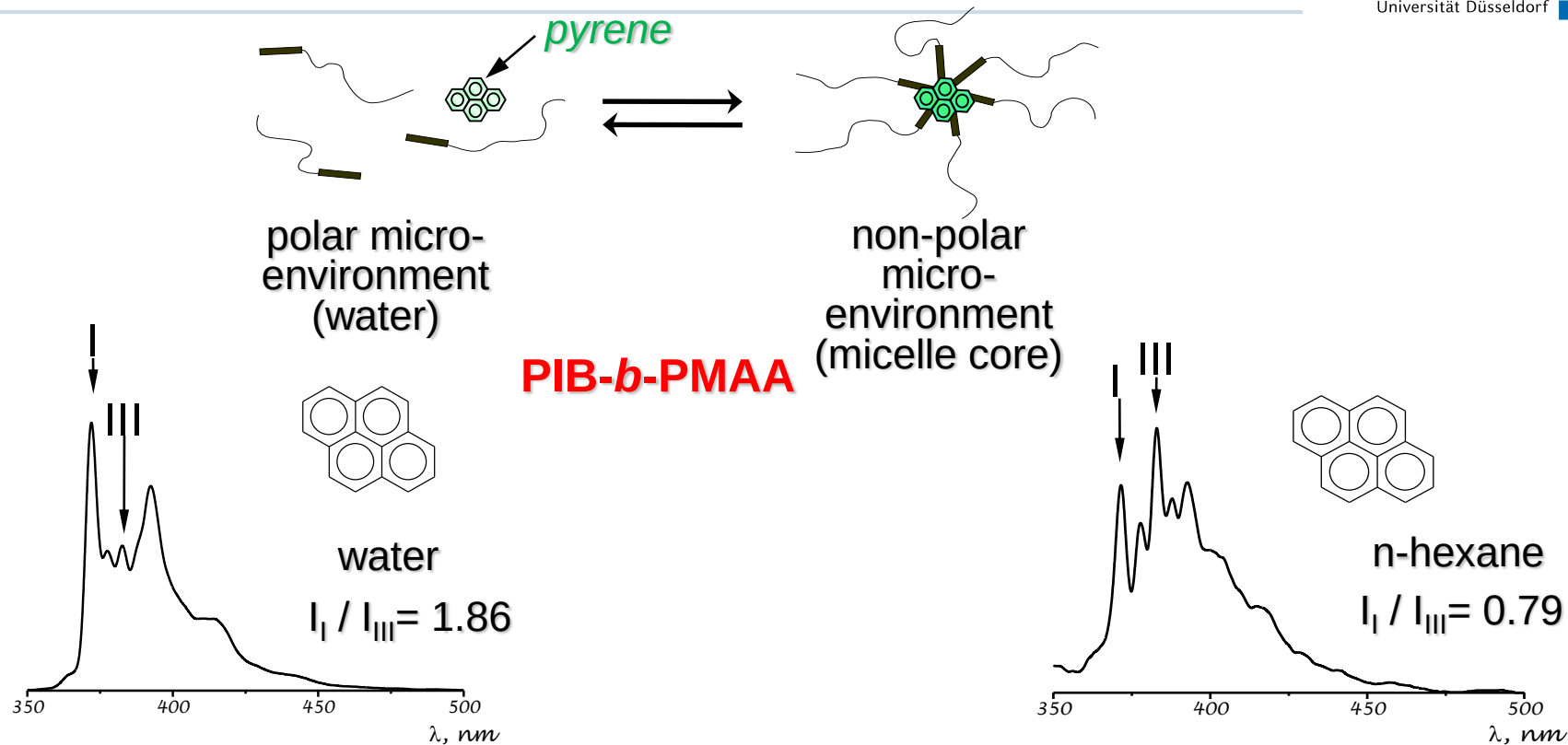
Regular arrangement due
to electrostatic repulsion

Statistics of ca. 40 particles:
Distance to the 6 next
neighbours.

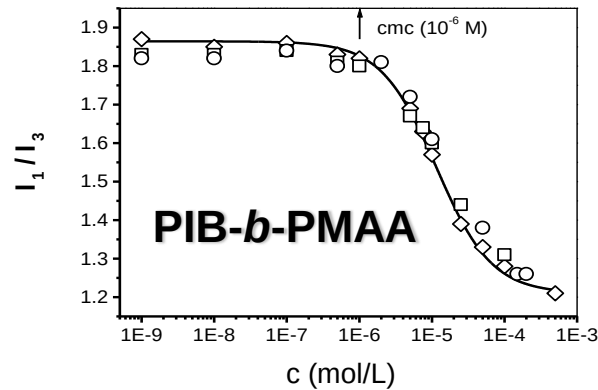
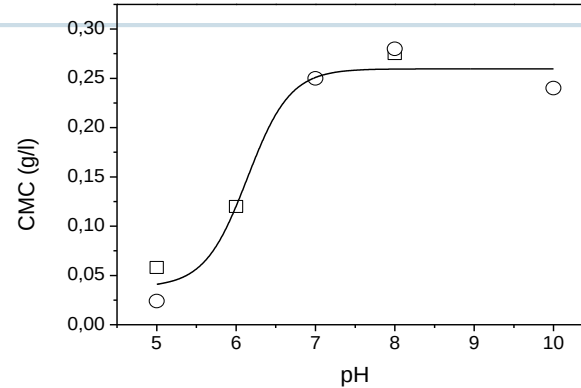
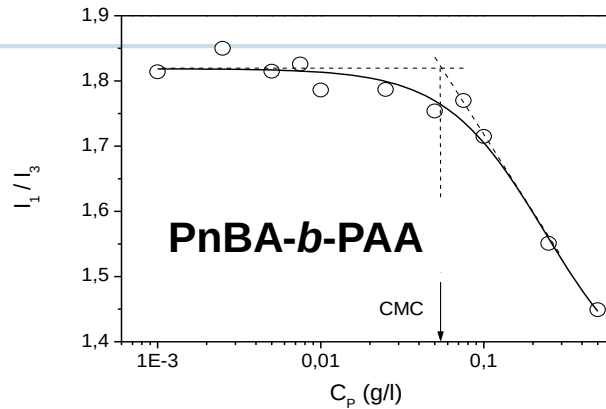
$$\langle d_{\text{app}} \rangle = 59 \text{ nm}$$

$c = 1 \text{ wt-}\%$; $\text{pH} \sim 9$

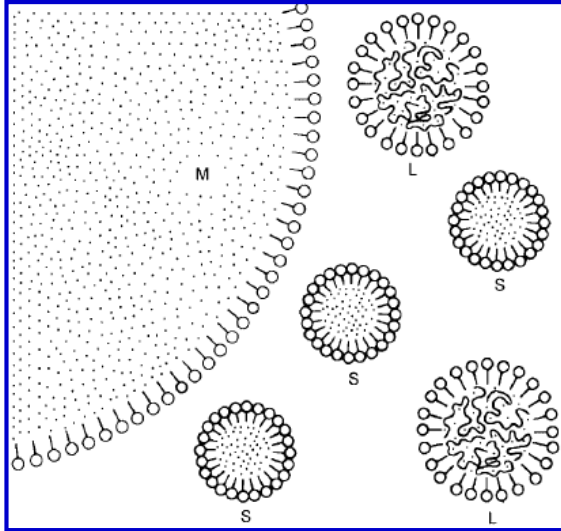
CMC determination by fluorometry



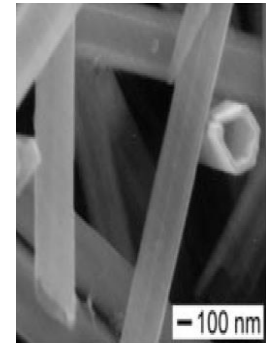
CMC determination by fluorometry



Some Applications of Block Copolymer Micelles and Vesicles



- Stabilizers in emulsion polymerization
- Lubricants in oil industry
- Drug and gene delivery
- Removal of pollutants from water
- Nanoreactors and nanotubes





- Synthese von Blockcopolymeren BCP
- Sequentielle Monomerzugabe
- Reaktivitäten der Monomerbausteine relevant
- Lebende Copolymerisation
- Phasendiagramm im Festkörper für Diblöcke
- Thermoplastische Elastomere (Triblöcke), ABA
- Blockcopolymere in Lösung
- Aggregation; Mizellenbildung
- Amphiphile Blockcopolymere
- Einsatzgebiete BCP in Lösung

